

A Study for Forecasting Solar Energy Power Generation

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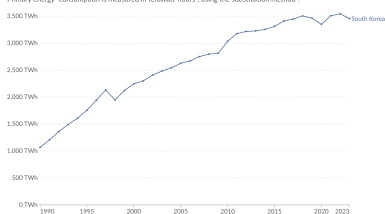
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Intro & Motivation

Introduction

Primary energy consumption

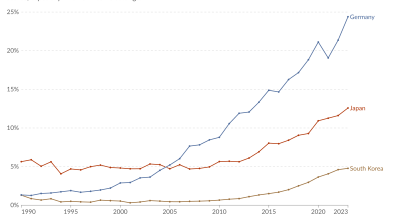
Primary energy¹ consumption is measured in terawatt-hours², using the substitution method³.



Data source: U.S. Energy Information Administration (2023); Energy Institute - Statistical Review of World Energy (2024)
Note: Data includes only commercially-traded fuels (coal, oil, gas), nuclear and modern renewables. It does not include traditional biomass.
OurWorldinData.org/energy | CC BY

Share of primary energy consumption from renewable sources

Measured as a percentage of primary energy¹ using the substitution method⁴. Renewables include hydropower, solar, wind, geothermal, bioenergy, wave, and tidal, but not traditional biofuels, which can be a key energy source, especially in lower-income settings.



Data source: Energy Institute - Statistical Review of World Energy (2024)
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- Korea's energy consumption continues to increase, and most of the materials for energy production are imported.
- Solar PV in major countries accounts for about 5% of Korea's total power generation, compared to 12% in Germany and 10% in Japan.
- Solar power generation is slowly increasing in the proportion of total power generation because solar power plants are continuously being installed across the country and the capacity of facilities is increasing.

Percentage of imports(https://www.index.go.kr/unity/potal/main/EachDtlPageDetail.do?idx_cd=1339)

Share in Korea (<https://www.iea.org/countries/korea/electricity>)

Germany (https://www.bundesnetzagentur.de/SharedDocs/Pressemitteilungen/EN/2024/20240103_SMARD)

Japan(<https://www.iea.org/data-and-statistics/data-tools/energy-statistics-data-browser?country=JAPAN&fuel=Energy%20supply&indicator=ElecGenByFuel>)

Facility capacity in Koera(https://recloud.energy.or.kr/m/present/present01_01.do)

Motivation

- If eco-friendly energy production can be accurately predicted, it will be possible to efficiently produce the amount generated by other production methods (nuclear power, thermal power, etc.).
- Even if methods such as ARIMA, LSTM, and Transformer are applied, there are cases where predictions are not well made.
- So, We can apply the latest time series data prediction model CNN-LSTM-Transformer.

Data Description

Data Sources

- They are all hourly data and the region is Seoul.

1. Solar data

- Solar-power data were obtained from KEPCO's *Time-Series Solar & Wind Generation by Region* dataset (Korea public-data portal).
- Solar power generation data is the transaction volume of the electricity market and is different from the actual power generation.

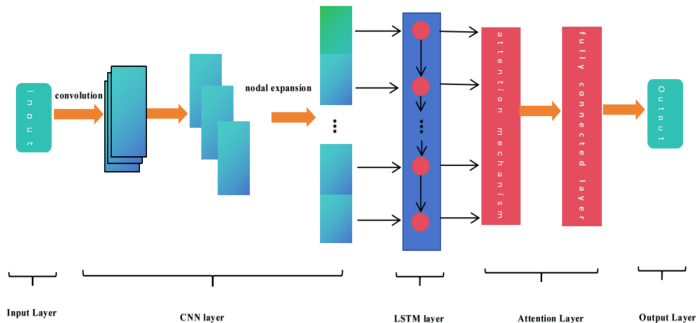
2. Weather data

- Weather data were obtained from the *Automated Synoptic Observing System (ASOS)* via the KMA Meteorological Data Portal.
- Ground temperature, temperature, precipitation, wind speed, wind direction, humidity, air pressure, etc.

Meteorological Data Open Portal, Automated Surface Observing System (ASOS)
(<https://data.kma.go.kr/data/grnd/selectAsosRltmList.do?pgmNo=36>)
Korea Power Exchange, Regional Hourly Solar and Wind Power Generation
(<https://www.data.go.kr/data/15065269/fileData.do?recommendDataYn=Y>)

Methods

hybrid CNN-LSTM-Transformer



- CNN: Extracts local temporal and crossvariable patterns via causal 1D convolutions.
- LSTM: Learns sequential dependencies and retains longshort-term information through gated recurrent units.
- Transformer: Longrange dependencies using multihead selfattention.

Tian, A. (2024). Enhancing vegetable sales forecasting with a CNN-LSTM-Transformer hybrid model. Highlights in Business, Economics and Management, 25, 112121. <https://doi.org/10.54097/rr7t4d15>

Al-Ali, E. M., et al. (2023). Solar energy production forecasting based on a hybrid CNN-LSTM-Transformer model. Mathematics, 11(3), 676. <https://doi.org/10.3390/math11030676>

Data Input

- Shape: (number of windows N , sequence length L , number of variables F).
- Each input is a multivariate time series window, min max scaled.

CNN Stage

- Two 1-D convolutional layers extract local temporal patterns and cross-variable correlations.
- A max pooling layer reduces the sequence length.
- Output shape: (N, L', C) , where L' is halved by pooling, and C is the number of filters.

LSTM Stage

- LSTM processes the CNN features to capture medium- and long-term dependencies.
- Output shape: (N, L', H) , where H is the hidden size.

Transformer Stage

- Transformer encoder layers apply multi-head self-attention to model global (long-range) patterns.
- Only encoder blocks are used; decoder blocks are excluded.
- Positional encoding is added to the LSTM output.
- Output shape: $(N, L' \text{ or } 1, D)$.

Output Stage

- Final timestep vector is passed through a fully connected layer.
- Returns predicting solar power generation.

Experiments

Experimental 1

- Input data period : 1 years ($365 \times 24 = 8760$ hours)
- Predict period : 1 days (24 hours)
- Comparing the RMSE of the models ARIMA, TimesFM, CNN-LSTM, and CNN-LSTM-TF.
- 100 times simulated and get mean of RMSE

	CNN-LSTM	CNN-LSTM-TF	ARIMA	TimesFM
RMSE	0.392	0.304	4.2	3.19

Lower values indicate better performance.

Need for Synthetic Data

Suppose the amount of data available for training is insufficient.

- A time series data version of Diffusion, a model that generates synthetic data, has been developed.
- Recently, studies have shown that synthetic time series made from time series diffusion models alleviate the data shortage problem of predictive models.

Yuan, X., et al. (2024). Diffusion-TS: Interpretable diffusion for general time-series generation. Paper presented at the 12th International Conference on Learning Representations (ICLR 2024)

Kollovieh, M., et al. (2023). Predict, refine, synthesize: Self-guiding diffusion models for probabilistic time-series forecasting (arXiv preprint arXiv:2307.11494). <https://doi.org/10.48550/arXiv.2307.11494>

Experimental 2

- Input data period : 2 weeks ($15 \times 24 = 360$ hours)
- Compare only use real data, stacked real data and synthetic data
- Predict period : 1 days (24 hours)
- Comparing the RMSE of the models ARIMA, Times FM, Transformer, CNN-LSTM, and CNN-LSTM-TF.

mean-RMSE	CNN-LSTM	CNN-LSTM-TF	ARIMA	TimesFM
Real-data	2.906	2.968	4.195	2.304
Stacked-data	1.972	1.995		

Lower values indicate better performance.

Thank you for listening